

Prescribed-Time Attitude Tracking Control of Rigid Spacecraft with Guaranteed Performance under Arbitrary Initial Conditions

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Abstract. This study examines the prescribed-time attitude tracking control problem for rigid spacecraft with guaranteed performance under arbitrary initial conditions. A smooth time-varying gain is first constructed to establish a practical prescribed-time stability criterion. Then, an improved prescribed performance function (PPF) is devised by using the initial value and sign of the tracking error, so that the conventional initial-bound restriction can be removed. Based on the transformed error, a prescribed-time adaptive attitude tracking controller is devised to compensate for system uncertainties and ensure convergence within a prescribed time. Comparative simulations under multiple initial conditions validate the feasibility and robustness.

Keywords: Prescribed-time control, adaptive fuzzy control, attitude tracking, performance-constrained control method

1 Introduction

With the continuous advancement of modern aerospace missions, increasingly higher demands have been imposed on the convergence performance and tracking accuracy of attitude control systems [1]. Prior to practical deployment, it is essential to conduct verifiable performance evaluations of control algorithms to ensure that the desired specifications can be achieved. To this end, such algorithms are required to possess quantitative performance stability [2], enabling analytical characterization of the upper bounds on settling time and residual error.

Fixed-time and prescribed-time control methods are effective tools for improving convergence performance. For both approaches, the settling time admits a uniform upper bound and unaffected by the initial conditions [3, 4]. However, in fixed-time control, the association between the convergence-time bound and controller gains is typically complicated, making parameter tuning more challenging. In contrast, prescribed-time control explicitly parameterizes the convergence-time bound within the controller design, thereby significantly simplifying the controller synthesis process [5]. In recent years, this approach has attracted considerable attention in spacecraft attitude control applications [6, 7]. However, most existing prescribed-time attitude control approaches rely on sliding mode control, which may lead to chattering issues.

In order to improve transient response, a PPF was introduced in [8], which enforces predefined constraints on the tracking error and achieves unified control of transient and steady-state performance. Motivated by these developments, prescribed performance techniques have been broadly utilized in spacecraft attitude control [9, 10]. Nevertheless, many existing approaches require the initial tracking error to satisfy the prescribed constraint region. This requirement typically necessitates manual tuning of boundary parameters for specific application scenarios, which not only increases implementation complexity but also limits the generality and practical applicability of the method.

To address these problems, a prescribed-time attitude tracking control strategy is proposed for rigid spacecraft. This work makes the following key contributions. First, a smooth time-varying gain is introduced to derive a practical prescribed-time stability condition. Second, the initial tracking error and its sign are embedded into a redesigned performance function, eliminating the requirement that the initial error lies within predefined bounds. Third, a prescribed-time adaptive attitude tracking controller is designed to compensate for system nonlinearities while ensuring prescribed-time convergence.

2 Problem Formulation and Preliminaries

2.1 Dynamic Modeling of a Rigid Spacecraft

The rotational kinematics and dynamics of a rigid spacecraft are represented below in the framework of Modified Rodrigues Parameters (MRPs) [1]

$$\dot{\rho} = H(\rho)\nu \quad (1)$$

$$M\dot{\nu} = -\nu^\times M\nu + \tau_u + \tau_d \quad (2)$$

where $\rho = [\rho_1, \rho_2, \rho_3]^T \in \mathbb{R}^3$ denotes the spacecraft attitude relative to the inertial frame; $H(\rho) \in \mathbb{R}^{3 \times 3}$ is given by $H(\rho) = \frac{1}{4}[(1 - \rho^T \rho)E_3 + 2\rho^\times + 2\rho\rho^T]$; $M \in \mathbb{R}^{3 \times 3}$ is the inertia matrix; $\nu = [\nu_1, \nu_2, \nu_3]^T \in \mathbb{R}^3$ stands for the angular velocity; $\tau_u = [\tau_{u1}, \tau_{u2}, \tau_{u3}]^T \in \mathbb{R}^3$ stands for the control torque; $\tau_d = [\tau_{d1}, \tau_{d2}, \tau_{d3}]^T \in \mathbb{R}^3$ stands for the external disturbance satisfying $\|\tau_d\| \leq \bar{\tau}_d$, where $\bar{\tau}_d$ is an unknown constant.

Since $M = M_0 + \Delta M$, (2) can be rewritten as

$$M_0\dot{\nu} = -\nu^\times M\nu - \Delta M\dot{\nu} + \tau_u + \tau_d \quad (3)$$

Consider the attitude tracking problem, where the attitude error and angular velocity error are defined as $e = [e_1, e_2, e_3]^T \in \mathbb{R}^3$ and $v_e = [v_{e1}, v_{e2}, v_{e3}]^T \in \mathbb{R}^3$, respectively

$$e = \rho \otimes \rho_r^{-1} = \frac{\rho_r(\rho^T \rho_r - 1) + \rho(1 - \rho_r^T \rho_r) - 2\rho_r^\times}{1 + \rho_r^T \rho_r \rho^T \rho + 2\rho_r^T \rho} \quad (4)$$

$$v_e = \nu - R\nu_r \quad (5)$$

where $\rho_r = [\rho_{r1}, \rho_{r2}, \rho_{r3}]^T \in \mathbb{R}^3$ and $\nu_r = [\nu_{r1}, \nu_{r2}, \nu_{r3}]^T \in \mathbb{R}^3$ denote the reference attitude and angular velocity, respectively, while R denotes the direction cosine matrix corresponding to the attitude error.

Based on (1)-(5), the dynamics of the attitude error and angular velocity error can be derived as

$$\dot{e} = H(e)v_e \quad (6)$$

$$M_0\dot{v}_e = -v^\times Mv - \Delta M\dot{v} - M_0 R\dot{v}_r + M_0 v_e^\times Rv_r + \tau_u + \tau_d \quad (7)$$

By defining $y_1 = e$ and $y_2 = \dot{e}$, the spacecraft attitude tracking error system models (6) and (7) can be reformulated as follows

$$\begin{cases} \dot{y}_1 = y_2 \\ \dot{y}_2 = \mathcal{F} + \mathcal{L} + H(y_1)M_0^{-1}\tau_u + H(y_1)M_0^{-1}\tau_d \end{cases} \quad (8)$$

where $\mathcal{F} = [\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3]^T = H(y_1)v_e - H(y_1)R\dot{v}_r - H(y_1)M_0^{-1}v^\times M_0v_e + H(y_1)v_e^\times Rv_r$, $\mathcal{L} = [\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3]^T = H(y_1)M_0^{-1}(-v^\times \Delta Mv - \Delta M\dot{v})$.

2.2 Practical Prescribed-Time Stability

Definition 1. Consider the nonlinear system

$$\dot{y} = \varphi(y, \tau), \quad \varphi(0, \tau) = 0 \quad (9)$$

where $\varphi : \mathbb{R}^p \times \mathbb{R}^+ \rightarrow \mathbb{R}^p$ is continuous and nonlinear, and $y = 0$ is an equilibrium point.

The system is prescribed-time stable (PTS) if there exists $T_p > 0$ such that $y(\tau) = 0$ for all $\tau \geq T_p$. It is practically prescribed-time stable (PPTS) if there exist $T_p > 0$ and $\epsilon > 0$ such that $\|y(\tau)\| \leq \epsilon$ for all $\tau \geq T_p$.

First, a new time-varying gain function $\lambda(\tau)$ is given as follows:

$$\lambda(\tau) = \begin{cases} \frac{1}{\epsilon} \sin^2\left(\frac{\pi\tau}{2T_p}\right), & \tau \in [0, T_p] \\ \frac{1}{\epsilon}, & \tau \in (T_p, +\infty) \end{cases} \quad (10)$$

where $T_p > 0$ and $\epsilon > 0$ are design parameters satisfying $0 < \epsilon \ll 1$. $\lambda(\tau)$ has the following characteristics:

1. $\lambda(\tau)$ is continuous, nondecreasing on $[0, +\infty)$, with $\lambda(0) = 0$ and $\lambda(T_p) = 1/\epsilon$;
2. $\lambda(\tau) \in \mathbb{R}^1([0, +\infty))$.

Lemma 1. [11] Let $U(\tau)$ be a positive definite and radially unbounded Lyapunov function. Suppose that for some $a > 0$ and $\phi > 0$,

$$\dot{U}(\tau) \leq -a\lambda(\tau)U(\tau) + \phi \quad (11)$$

where $\lambda(\tau)$ is a time-varying gain function. Then, the system satisfies PPTS with T_p being the prescribed convergence time.

3 Main Results

This section proposes the prescribed performance transformation, the prescribed-time backstepping controller, and the associated stability analysis.

3.1 Improved PPF Design

For prescribed performance assurance, the attitude tracking error $e_j(\tau)$ satisfies the following inequality:

$$\underline{\chi}(\tau) < e_j(\tau) < \bar{\chi}(\tau) \quad (12)$$

where $\underline{\chi}(\tau)$ and $\bar{\chi}(\tau)$ are the improved PPFs, given by

$$\begin{cases} \bar{\chi}(\tau) = \text{sign}(e_j(0))(\phi(\tau) - \phi_\infty) + \bar{\varrho}\phi(\tau) \\ \underline{\chi}(\tau) = \text{sign}(e_j(0))(\phi(\tau) - \phi_\infty) - \underline{\varrho}\phi(\tau) \end{cases} \quad (13)$$

with

$$\phi(\tau) = \begin{cases} \phi_\infty + (|e_j(0)| + \varpi) \left(1 - \frac{\tau}{T_r}\right)^p, & 0 \leq \tau < T_r \\ \phi_\infty, & \tau \geq T_r \end{cases} \quad (14)$$

where $e_j(0)$, $j = 1, 2, 3$, denote the initial attitude tracking errors; $0 < \bar{\varrho} < 1$, $0 < \underline{\varrho} < 1$, $\phi_\infty > 0$, and $\varpi > 0$ are design parameters; T_r denotes the convergence time constant, while $p > 0$ is used to tune the smoothness of the boundary function.

Then, differentiating (13) and (14) results in

$$\begin{cases} \dot{\bar{\chi}}(\tau) = \left[\text{sign}(e_j(0)) + \bar{\varrho} \right] \dot{\phi}(\tau) \\ \dot{\underline{\chi}}(\tau) = \left[\text{sign}(e_j(0)) - \underline{\varrho} \right] \dot{\phi}(\tau) \end{cases} \quad (15)$$

and

$$\dot{\phi}(\tau) = \begin{cases} -\frac{p}{T_r} (|e_j(0)| + \varpi) \left(1 - \frac{\tau}{T_r}\right)^{p-1}, & 0 \leq \tau < T_r \\ 0, & \tau \geq T_r \end{cases} \quad (16)$$

According to (14) and (16), both $\phi(\tau)$ and its time derivative $\dot{\phi}(\tau)$ have identical left- and right-hand limits as $\tau \rightarrow T_r$, i.e., $\lim_{\tau \rightarrow T_r^-} \phi(\tau) = \lim_{\tau \rightarrow T_r^+} \phi(\tau) = \phi_\infty$ and $\lim_{\tau \rightarrow T_r^-} \dot{\phi}(\tau) = \lim_{\tau \rightarrow T_r^+} \dot{\phi}(\tau) = 0$. This implies that $\phi(\tau)$ is continuously differentiable at $\tau = T_r$, i.e., $\phi(\tau) \in \mathbb{R}^1$.

The constrained error condition in (12) is converted into an unconstrained form by introducing the following transformed error:

$$\psi_j = \ln \left(\frac{\vartheta_j}{1 - \vartheta_j} \right) \quad (17)$$

where ϑ_j is defined as $\vartheta_j = \frac{e_j - \underline{\chi}}{\bar{\chi} - \underline{\chi}}$, and the inverse function of (17) is given by

$$e_j = S(\psi_j) = \frac{\underline{\chi}(\tau) + \exp(\psi_j)\bar{\chi}(\tau)}{1 + \exp(\psi_j)} \quad (18)$$

where $S(\psi_j)$ is a continuously differentiable and strictly increasing function satisfying the following properties:

$$\begin{cases} \underline{\chi}(\tau) < S(\psi_j) < \bar{\chi}(\tau), \forall \tau \geq 0 \\ \lim_{\psi_j \rightarrow +\infty} S(\psi_j) = \bar{\chi}, \lim_{\psi_j \rightarrow -\infty} S(\psi_j) = \underline{\chi} \end{cases} \quad (19)$$

3.2 Controller Design

To achieve prescribed transient and steady-state performance, the transformed error in (17) is adopted in the controller design. The prescribed-time control design is carried out as shown below:

Step 1: Differentiating ϑ_j is derived as

$$\dot{\vartheta}_j = \frac{\dot{e}_j [\bar{\chi}(\tau) - \underline{\chi}(\tau)] - \dot{\chi}(\tau) [\bar{\chi}(\tau) - e_j] - \dot{\bar{\chi}}(\tau) [e_j - \underline{\chi}(\tau)]}{[\bar{\chi}(\tau) - \underline{\chi}(\tau)]^2} \quad (20)$$

Differentiating (17) yields

$$\dot{\psi}_j = \frac{\dot{\vartheta}_j}{\vartheta_j(1 - \vartheta_j)}. \quad (21)$$

Substituting (20) into (21) leads to

$$\begin{aligned} \dot{\psi}_j &= \left[\frac{1}{e_j - \underline{\chi}(\tau)} + \frac{1}{\underline{\chi}(\tau) - e_j} \right] \dot{e}_j - \frac{\dot{\chi}(\tau)}{e_j - \underline{\chi}(\tau)} - \frac{\dot{\bar{\chi}}(\tau)}{\bar{\chi}(\tau) - e_j} \\ &= r_j \dot{e}_j - b_j \end{aligned} \quad (22)$$

where $r_j = \left[\frac{1}{e_j - \underline{\chi}(\tau)} + \frac{1}{\bar{\chi}(\tau) - e_j} \right]$, $b_j = \frac{\dot{\chi}(\tau)}{e_j - \underline{\chi}(\tau)} + \frac{\dot{\bar{\chi}}(\tau)}{\bar{\chi}(\tau) - e_j}$, $j = 1, 2, 3$.

The virtual error is then defined as $s = y_2 - v$, where $v = [v_1, v_2, v_3]^T$ denotes the virtual controller, given by

$$v = -r^{-1}[a_1 \lambda(\tau) \psi - b] \quad (23)$$

where a_1 is a positive constant, $r = \text{diag}\{r_1, r_2, r_3\}$, $\psi = [\psi_1, \psi_2, \psi_3]^T$, $b = [b_1, b_2, b_3]^T$, and $\lambda(\tau)$ is given by

$$\lambda(\tau) = \begin{cases} \frac{1}{\epsilon} \sin^2\left(\frac{\pi\tau}{2T_r}\right), & \tau \in [0, T_r] \\ \frac{1}{\epsilon}, & \tau \in (T_r, +\infty) \end{cases} \quad (24)$$

where T_r and ϵ are positive constants, and $0 < \epsilon \ll 1$.

Substituting (23) into (22) leads to

$$\dot{\psi} = r s - a_1 \lambda(\tau) \psi \quad (25)$$

Step 2: Differentiating the virtual error s becomes

$$\dot{s} = \dot{y}_2 - \dot{v} \quad (26)$$

According to the spacecraft system model (8), $\dot{s}$ can be further rewritten as

$$\begin{aligned} \dot{s} &= \mathcal{F} + \mathcal{L} + H(y_1)M_0^{-1}\tau_u + H(y_1)M_0^{-1}\tau_d - \dot{v} \\ &= \Pi + \mathcal{F} + H(y_1)M_0^{-1}\tau_u + H(y_1)M_0^{-1}\tau_d \end{aligned} \quad (27)$$

where $\Pi = [\Pi_1, \Pi_2, \Pi_3]^T = \mathcal{L} - \dot{v}$ represents the total system uncertainty. To handle this uncertainty, a fuzzy logic system (FLS) is utilized [12], whose approximation for the j -th component can be written as

$$\Pi_j = A_j^T \Psi_j(\iota_j) + \varsigma_j, j = 1, 2, 3 \quad (28)$$

where $\iota_j = [y_{1j}, y_{2j}, v_j]^T \in \mathbb{R}^3$ denotes the input vector of the fuzzy logic system; $A_j \in \mathbb{R}^3$ denotes the weight vector; $\Psi_j(\cdot)$ is the basis function; and ς_j represents the approximation error, which is bounded by $|\varsigma_j| \leq \varsigma_N$, where ς_N is an unknown positive constant.

The prescribed-time attitude tracking controller $\tau_u = [\tau_{u1}, \tau_{u2}, \tau_{u3}]^T$ is devised as

$$\tau_u = -M_0 H(y_1)^{-1} \left[\mathcal{F} + a_2 \lambda(\tau) s + r\psi + \frac{\hat{\Theta} \Omega s}{2\ell^2} \right] \quad (29)$$

where $\Omega = \text{diag}\{\Psi_1^T \Psi_1, \Psi_2^T \Psi_2, \Psi_3^T \Psi_3\}$, $\Theta = \max\{\|A_1\|^2, \|A_2\|^2, \|A_3\|^2\}$, $a_2 = 1 + a'_2$, $a'_2 > 0$, $\ell > 0$, and $\hat{\Theta}$ is the estimation of Θ .

Substituting (29) into (27) yields

$$\dot{s} = \Pi - a_2 \lambda(\tau) s - r\psi + H(y_1) M_0^{-1} \tau_d - \frac{\hat{\Theta} \Omega s}{2\ell^2}. \quad (30)$$

The adaptive updating law of $\hat{\Theta}$ is formulated as

$$\dot{\hat{\Theta}} = \frac{\sum_{j=1}^3 s_j^2 \Psi_j^T \Psi_j}{2\ell^2} - a_3 \lambda(\tau) \hat{\Theta} \quad (31)$$

where a_3 is a positive constant.

3.3 Stability analysis

Theorem 1. *With the virtual controller (23), the prescribed-time attitude tracking controller (29), and the adaptive law (31) in place, the rigid spacecraft system (8) guarantees boundedness of all closed-loop signals. In addition, the attitude tracking errors e_j , $j = 1, 2, 3$, are driven to the prescribed residual set in (12) within the prescribed time T_r .*

Proof. Step 1: Choose a positive Lyapunov function as

$$U_1 = \frac{1}{2} \psi^T \psi + \frac{1}{2} s^T s. \quad (32)$$

Differentiating (32) results in

$$\dot{U}_1 = \psi^T \dot{\psi} + s^T \dot{s}. \quad (33)$$

Substituting (25) and (30) into (33) yields

$$\begin{aligned}\dot{U}_1 &= \psi(rs - a_1\lambda(\tau)\psi) + s\left(\Pi - a_2\lambda(\tau)s - r\psi + H(y_1)M_0^{-1}\tau_d - \frac{\hat{\theta}\Omega s}{2\ell^2}\right) \\ &= -a_1\lambda(\tau)\psi^T\psi - a_2\lambda(\tau)s^2 + s^T\mathcal{P} + s^TH(y_1)M_0^{-1}\tau_d - \frac{\hat{\theta}\sum_{j=1}^3 s_j^2\Psi_j^T\Psi_j}{2\ell^2}.\end{aligned}\quad (34)$$

By utilizing Young's inequality, we have

$$\begin{aligned}s^T\Pi &\leq \sum_{j=1}^3 s_j A_j^T\Psi_j + \sum_{j=1}^3 s_j s_N \\ &\leq \frac{\Theta\sum_{j=1}^3 s_j^2\Psi_j^T\Psi_j}{2q^2} + \frac{3\ell^2}{2} + \frac{\|s\|^2}{2} + \frac{3s_N^2}{2},\end{aligned}\quad (35)$$

and

$$s^TH(y_1)M_0^{-1}\tau_d \leq \frac{\|s\|^2}{2} + \frac{\bar{\tau}_d^2\|M_0\|^2}{8}.\quad (36)$$

Substituting (35) and (36) into (34) yields

$$\begin{aligned}\dot{U}_1 &\leq -a_1\lambda(\tau)\|\psi\|^2 - a_2\lambda(\tau)\|s\|^2 + \|s\|^2 + \frac{3s_N^2}{2} + \frac{3\ell^2}{2} + \\ &\quad \frac{\bar{\tau}_d^2\|M_0\|^2}{8} + \frac{(\Theta - \hat{\theta})\sum_{j=1}^3 s_j^2\Psi_j^T\Psi_j}{2\ell^2} \\ &\leq -a_1\lambda(\tau)\|\psi\|^2 - a'_2\lambda(\tau)\|s\|^2 + \mathcal{D}_1 + \frac{(\Theta - \hat{\theta})\sum_{j=1}^3 s_j^2\Psi_j^T\Psi_j}{2\ell^2}\end{aligned}\quad (37)$$

where $\mathcal{D}_1 = \frac{3s_N^2}{2} + \frac{3\ell^2}{2} + \frac{\bar{\tau}_d^2\|M_0\|^2}{8}$.

Step 2: Choose a positive Lyapunov function as

$$U_2 = U_1 + \frac{1}{2}\tilde{\theta}^2\quad (38)$$

where $\tilde{\theta} = \theta - \hat{\theta}$.

Differentiating (38) results in

$$\dot{U}_2 = \dot{U}_1 - \tilde{\theta}\dot{\hat{\theta}}.\quad (39)$$

By substituting (31) into (39), it follows that

$$\dot{U}_2 = -a_1\lambda(\tau)\|\psi\|^2 - a'_2\lambda(\tau)\|s\|^2 + \mathcal{D}_1 + a_3\lambda(\tau)\tilde{\theta}\hat{\theta}.\quad (40)$$

By utilizing Young's inequality, one can obtain

$$\begin{aligned}a_3\lambda(\tau)\tilde{\theta}\hat{\theta} &= a_3\lambda(\tau)\tilde{\theta}(\theta - \tilde{\theta}) \\ &\leq -\frac{a_3\lambda(\tau)\tilde{\theta}^2}{2} + \frac{a_3\lambda(\tau)\theta^2}{2}.\end{aligned}\quad (41)$$

By substituting (41) into (42), it follows that

$$\begin{aligned}
\dot{U}_2 &\leq -a_1\lambda(\tau)\|\psi\|^2 - a'_2\lambda(\tau)\|s\|^2 - \frac{a_3\lambda(\tau)\tilde{\Theta}^2}{2} + \frac{a_3\lambda(\tau)\Theta^2}{2} + \mathcal{D}_1 \\
&\leq -a_1\lambda(\tau)\|\psi\|^2 - a'_2\lambda(\tau)\|s\|^2 - \frac{a_3\lambda(\tau)\tilde{\Theta}^2}{2} + \frac{a_3\Theta^2}{2\epsilon} + \mathcal{D}_1 \\
&= -a_4\lambda(\tau)U_2 + \mathcal{D}_2
\end{aligned} \tag{42}$$

where $a_4 = \min\{a_1, a'_2, \frac{1}{2}a_3\}$; $\mathcal{D}_2 = \frac{a_3\Theta^2}{2\epsilon} + \mathcal{D}_1$.

From Lemma 1, by selecting suitable constants a_4 and $\mathcal{D}_2$, the whole system achieves PPTS, and the attitude tracking error e enters a small residual set around the origin within the prescribed time T_r .

This completes the proof of Theorem 1.

4 Results of Simulations

To verify the independence characteristic with respect to initial conditions, two different initial attitudes are considered, namely (1) $\rho(0) = [-0.2, 0.4, 0.6]^T$ and (2) $\rho(0) = [0.5, -0.8, 1.2]^T$. For each case, two prescribed convergence times are selected as $T_r = 3$ s and $T_r = 5$ s. For the spacecraft system (1)-(2), the inertia matrix is set to $M_0 = \text{diag}\{1, 0.63, 0.85\}$, while the total disturbance (inertia uncertainties and external disturbances) are given by $\Delta M = \text{diag}\{\sin(0.15\tau), 2\sin(0.25\tau), 3\sin(0.35\tau)\}$ kg · m² and $\tau_d = [0.25\sin(0.15\tau), 0.35\sin(0.25\tau), 0.55\sin(0.25\tau)]^T$, respectively. The initial angular velocity is chosen as $v(0) = [0, 0, 0]^T$ rad/s, and the reference angular velocity is defined as $v_r = [0.15\sin(\tau/45), -0.21\sin(\tau/55), 0.25\cos(\tau/45)]^T$ rad/s.

The parameters associated with the virtual controller (23), prescribed-time controller (29), and adaptive updating law (31) are chosen as $a_1 = a_2 = a_3 = 2$, $\ell = 0.1$, $\epsilon = 0.01$, and $\hat{\Theta}(0) = 0.1$, respectively. Moreover, the parameters of the improved PPFs (15) are specified as $\varpi = 0.01$, $\phi_\infty = 0.02$, $\bar{\varrho} = \underline{\varrho} = 0.3$, and $p = 1$.

The other results are given in Figs. 1–4. As can be observed, the spacecraft attitude tracking errors enter a small residual set around the origin with high accuracy within the prescribed time, regardless of the initial conditions. Moreover, the entire convergence process remains never exceeds the prescribed performance thresholds, achieving a steady-state accuracy better than 0.008. Specifically, under the same user-defined time (Figs. 1 and 3, as well as Figs. 2 and 4), both initial attitude cases (1) and (2) exhibit similar performance. The proposed PPF adaptively modifies its initial boundary according to the magnitude of the initial error, without requiring any redesign.

Results indicate that the proposed approach can ensure that the tracking error converges within a specified time frame. The initial performance envelope is adapted automatically according to the size of the initial error. Since the steady-state precision, the convergence duration, and the overshoot are all explicitly adjustable, the controller becomes more valuable in practice.

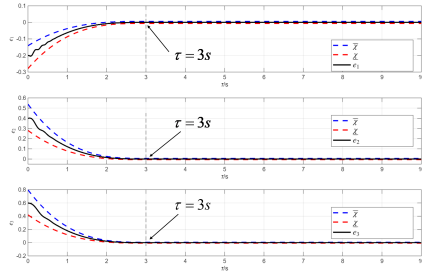


Fig. 1: Attitude tracking error curve under initial condition (1) and $\tau=3s$.

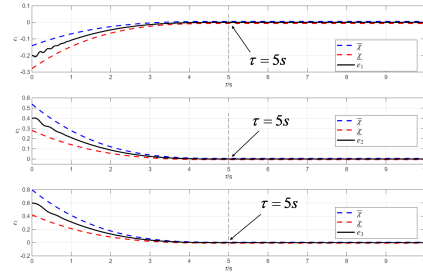


Fig. 2: Attitude tracking error curve under initial condition (1) and $\tau=5s$.

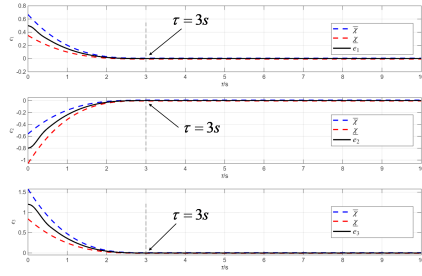


Fig. 3: Attitude tracking error curve under initial condition (2) and $\tau=3s$.

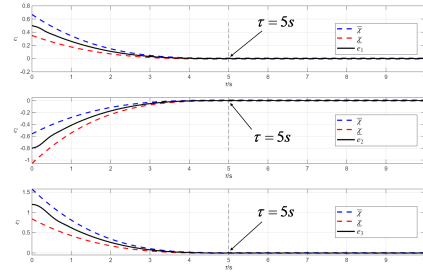


Fig. 4: Attitude tracking error curve under initial condition (2) and $\tau=5s$.

5 Conclusion

This work addresses adaptive prescribed-time attitude tracking control problem for rigid spacecraft under arbitrary initial conditions. The proposed framework is composed of three main components. First, a practical prescribed-time stability criterion is established via a time-varying gain function, ensuring convergence within a prescribed time. Second, improved PPFs are constructed by incorporating the initial values and signs of the tracking errors, enabling regulation of both transient and steady-state performance without requiring initial-bound constraints. Third, a prescribed-time adaptive attitude tracking controller is developed to compensate for system uncertainties and achieve fast and accurate tracking performance. Simulation results under different initial conditions verify the effectiveness and superiority of the proposed control framework.

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